

Reduced Food Neophobia and Food Disgust in Colorblind Men

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Author Note

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Abstract

Food neophobia is the fear of new foods, and it is associated with negative health outcomes. Recent work suggests that it is negatively correlated with the ability to discriminate visually similar dishes, but only in color. This led to a novel prediction: that colorblindness would be negatively associated with food neophobia. Here, we replicated that colorblind men ($n=103$) are less food neophobic than non-colorblind men ($n=273$), even when participants are unaware that being colorblind or not was relevant to their selection. We extend these results to food disgust, and find that both food neophobia and food disgust increase with age in colorblind men, whereas they are stable in noncolorblind men. These results underscore the role of color perception on affective attitudes towards food, in the absence of demand characteristics or manipulation of color in images. Our results advance our understanding of non-perceptual and affective consequences of colorblindness. They also have possible implications for improving treatment approaches in cases of severe food neophobia.

Introduction

Food Neophobia (FN) is often defined as the fear of new, or novel foods (Pliner & Hobden, 1992; Rabadán & Bernabéu, 2021). FN can manifest in the avoidance of newly introduced flavors or unknown consistency of food (Lobos & Januszewicz, 2019), but also predicts the reluctance to try other novel foods such as organic or genetically modified foods (Schickenberg et al., 2008). This avoidance is associated with poorer dietary quality, metabolic risk factors, and increased risk of disease outcome (Rabadán & Bernabéu, 2021; Sarin et al., 2019). There is a growing body of research aimed at delineating associated characteristics of FN that may inform intervention efforts. One idea is that fundamental characteristics of FN may include "cognitive factors" such as food perception, mental representations, categorization of food items (Lafraire et al., 2016). Consistent with this view, FN is negatively correlated with the ability to discriminate between visually similar prepared foods (e.g., a roast and a meatloaf, or two different noodle dishes; Gauthier & Fiestan, 2022). In a recent study, Sun and Gauthier (2023) found that this correlation between food-recognition ability and FN was specific to colored images of prepared food. This led to the prediction that colorblind people (men, in practice, because color blindness is much more prevalent in men; Xie et al., 2014) could have lower FN, because of a reduced range in color.

Because the relationships between color blindness and negative affective responses to foods that we uncover in this work are relatively novel, it is important to understand the original motivation for our predictions. They arose from recent efforts to explore visual ability to recognize food, as measured by discrimination of visually similar dishes (Gauthier & Fiestan, 2022). In that work, new measures for food recognition ability were developed and used to demonstrate evidence for a food-specific recognition ability. The logic was that different visual

tasks with food shared common variance even after controlling for domain-general object recognition. In that study, we explored what aspects of personality and experience can predict the ability to recognize food, and we found a negative relationship with FN. We proposed that this could be the result of experience: that people who avoid new foods have a relatively limited visual experience with food, resulting in limited perceptual expertise. A follow-up study (Sun & Gauthier, 2023) replicated these results, but with greyscale images of food. This work was, in turn, motivated by an emergent debate in the neuroimaging literature in trying to explain why the same regions of the ventral occipital system were selective for color and for images of food (Jain et al., 2023; Khosla et al., 2022; Pennock et al., 2022). By measuring performance on food tests without color present, we demonstrated that many aspects of visual food recognition did not depend on color. However, the negative relationship between food recognition and FN disappeared when color was absent (we found Bayesian support for a null effect). On this basis, Sun and Gauthier (2023) predicted that colorblind men, who experience a limited range in the color dimension, may experience less FN, and found support for the prediction. This was a novel prediction, in the absence of research linking aspects of visual experience and FN. An exception is a study that found no differences in FN for congenitally blind individuals and sighted individuals (Gagnon et al., 2013)– but the groups compared were small (13 vs. 15).

In the Sun & Gauthier (2023) study, sampling the population of colorblind men on the Prolific.co platform unexpectedly returned a response from men who were older than noncolorblind men ($M=41.4$ vs. 34.3). To address this challenge, the authors recruited additional older non-colorblind men to equate the age distributions. Here, we revisit the role of age as a possible modulator of the relation between FN and colorblindness, first in the Sun & Gauthier (2023) dataset and in the present study. FN shows a nonmonotonic dependency on age in

different studies, but a positive relationship with age in late adulthood (Hazley et al., 2022; van den Heuvel et al., 2019). FN also depends on education, social class, and living in an urban location (Rabadán & Bernabéu, 2021), and its relationship with age could reflect cohort effects and vary with cultural influences. Cultural influences on attitudes towards food can be complex and require extensive sampling efforts (e.g., Castro & Chambers, 2019). In contrast, colorblindness is a genetic condition that crosses all these other variables. Studying its effects could provide a window into some of the more universal roles of experience in FN. However, it is critically important to replicate the Sun and Gauthier results (2023) given that theirs is not only a novel finding, but the first result linking colorblindness to attitudes.

Although FN is characterized by the reluctance to eat and/or avoidance of novel foods, this avoidance may also be motivated by disgust. Indeed, disgust is believed to have evolved from a system of rejection of potential toxins in food (Rozin et al., 1997). Consistent with this view, both FN and pathogen disgust predict a reduced willingness to eat insects (White et al., 2023). Disgust and FN can both arise to protect organisms from eating potentially harmful foods, although they may have slightly different origins (Rozin et al., 1997). Both FD and FN predict the acceptance of cultured meat, a new food technology that promises to produce meat in a more environmentally friendly and animal-friendly way (Siegrist & Hartmann, 2020). Given their positive correlation (Thibodeau et al., 2021), FN and FD could be similarly influenced by colorblindness.

In addition, we measure personality because there is evidence of significant correlations between FN and disgust with personality traits. Although FN and FD are distinct constructs, the association between the two could be attributable to a latent personality variable (i.e., neuroticism, negative affect) that is common to all vulnerabilities. Studies (Knaapila et al., 2011;

Gauthier & Fiestan, 2023; Sun & Gauthier, 2023) have found negative correlations of FN with agreeableness and extraversion. Trait disgust correlates positively with neuroticism and conscientiousness, and negatively with agreeableness (Druschel & Sherman, 1999). In addition to their results with FN, Sun and Gauthier (2023) found that colorblind men were more agreeable and extraverted than noncolorblind men.

Re-analysis of Sun & Gauthier (2023)

We revisited the Sun & Gauthier (2023) results to investigate the effect of age on the results (data available at Gauthier, 2022). A multiple regression on FN scores included age (18-65), group (colorblind, normal color vision) and their interaction. While age itself was not a significant predictor, both groups (lower FN in colorblind men than in noncolorblind men, $p = .005$) and the interaction of age and group ($p = .03$) were significant predictors. Subsequent analysis probing this interaction showed that age was correlated with FN in colorblind men ($r = .26$, $p = .005$) while it was not in non-colorblind men ($r = -.03$, $p = .72$).

Hypotheses

The purpose of the current research is to examine the correlation between colorblindness, FN and FD. Based on the findings initially reported by Sun & Gauthier (2023), we hypothesized that FN (H1a) and FD (H1b) will be lower in colorblind men than in men with normal color perception. Based on our reanalysis of the role of age in that study, we hypothesized that age will moderate these effects for both FN (H2a) and FD (H2b), with a positive correlation of these variables with age expected for colorblind men but not for men with normal color vision.

To test for the specificity of these effects in the domain of food attitudes, we measured personality traits and expected the relationships specified in H1 and H2 not to be significant for personality traits (H3).

Study 1a

Method

Participants

Following Sun & Gauthier (2023), we aimed to sample a group of colorblind men on Prolific first, and match the age distribution for non-colorblind men. In March 2023, we made the task available to all men from the US on Prolific.co, who were fluent in English and answered “Yes, I’m colourblind” to the pre-screen question. The platform listed 370 matching participants active in the last 90 days. After a few days, once responses were no longer coming in, we closed the study, after collecting data from 107 volunteers. We excluded anyone who reported sex as 'female' or whose median RT on the Food Disgust Scale (FDS; the only measure for which we have RTs) were less than 500 ms (indicating poor attention to the survey). We were left with 103 participants. We recruited at least 2.5 the number of noncolorblind men to boost the power and precision of our comparisons. We collected data from 277 men in three groups according to the proportions above and after excluding those who reported sex as ‘female’ or whose median RT on the FDS was < 500 ms, 273 noncolorblind men remained. Demographic information is presented in Table 1. Differences between the two groups are consistent with the relative prevalence of colorblindness as a function of ethnicity (Xie et al., 2014). The study was approved by the local institutional review board.

Table 1. Group Demographics

	Colorblind	Non Colorblind
N	103	273
Age (mean (SD))	40.0 (14.1)	41.1 (14.6)
Ethnicity:		
White	79 (77%)	201 (74%)
Asian	12 (12%)	33 (12%)
Black		14 (5%)
Hispanic	9 (9%)	12 (4%)
Native American		4 (1%)
multiracial	3 (3%)	8 (3%)
other		1 (.4%)

Materials

The *Mini-IPIP* (Donnellan et al., 2006) is a 20-item scale, with four items measuring each of the five-factor model traits (Intellect/imagination (openness), conscientiousness, extraversion, agreeableness, and neuroticism). The participants responded to how well each statement (e.g., “I sympathise with others’ feelings”) described them using from 1 (not very well) to 5 (very well). Scores for individual items from each scale were averaged to produce a total score for each of the five scales.

The *Food Neophobia Scale* (FNS; Pliner & Hobden, 1992) consists of 10 statements such as "I am afraid to eat things I have never had before" and "I don't trust new foods," rated on a scale ranging from 1 (strongly disagree) to 7 (strongly agree). Five items on the scale were reverse coded, for instance, 'I will eat almost anything' and 'At dinner parties, I will try a new food.' Following standard practice, we use the sum score after correcting for reverse coding (on a 10-70 scale).

The *Food Disgust Scale* (FDS; Hartmann & Siegrist, 2018; Thibodeau et al., 2021) is a 32-item measure of disgust sensitivity to food and food-related situations. Participants are asked

to choose 'how grossed out they would be' for various situations. While the answers to the other two scales were collected in a way that did not record response times (with radio button-like scales), each item of the FDS was a sentence on the screen (e.g., 'If a friend bites into my bread' or 'To eat steak that is still bloody inside') with 6 buttons at the bottom of the screen for 1: 'not grossed out' to 6: 'extremely grossed out'. This version allowed response times to be collected. The mean of the median RTs on each of the 32 items was 2822 ms (SD = 1159). Following Thibodeau et al. (2021) we use the mean score over all items (on a scale of 1 to 6).

Procedure

All participants were tested online through Prolific, and began with a screen to ensure the task was not done on a mobile phone. One advantage of testing on Prolific is that asking about colorblindness is one of a large number of prescreen questions participants complete when they sign up on the platform. It is then possible to make one's study available to people who are colorblind (or not) without making them aware of why they are recruited, which eliminates possible concerns with demand characteristics. Participants completed the MINI-IPIP, the FNS, and the FDS in the same order.

Analyses

The hypotheses and the data analytic plan were specified prior to analyses of the data (which are available at Gauthier, 2023). The plan for Study 1b was conceived after collecting data for Study 1a, with the sole purpose of verifying whether the results changed if we measured colorblindness instead of using a self-report. Absolute values for skewness and kurtosis values

were less than one for all variables. Most analyses were performed in JASP 0.16.3 (JASP Team, 2023). Reliability was measured with Cronbach α and Guttman λ^2 (Callender & Osburn, 1979).

Results

Descriptive statistics

Table 2 presents descriptive statistics for all measures. Reliability was high for all measures.

Table 2. descriptive statistics and reliability

	Mean	SD	Skewness	Kurtosis	α	λ^2
FNS	28.258	14.05	0.71	0.126	.934	.936
FDS	3.622	0.882	0.042	0.126	.934	.938
Agreeableness	3.637	0.942	-0.426	0.126	.860	.862
Conscientiousness	3.765	0.839	-0.366	0.126	.760	.773
Extraversion	2.553	1.068	0.302	0.126	.877	.878
Intellect/Imagination	3.983	0.852	-0.769	0.126	.795	.803
Neuroticism	2.552	1.033	0.391	0.126	.822	.823

Zero-order correlations.

Table 3 shows the zero-order correlations between the study measures, with both groups combined. As expected, FNS and FDS were positively correlated. Table 3 also shows that FNS was negatively correlated with agreeableness, extraversion and imagination, and positively with neuroticism. FDS was positively correlated with only neuroticism.

Table 3. Pearson correlations

	FNS	FDS	A	C	E	I	N
FNS							
FDS	.39						
Agreeableness	-.23	.00					
Conscientiousness	-.08	.09	.19				
Extraversion	-.20	-.01	.43	.14			
Intellect/Imagination	-.18	-.07	.31	.16	.19		
Neuroticism	.12	.11	-.06	-.35	-.17	-.12	
Age	.10	.02	.00	.20	-.02	.02	-.17

note. r -values $> |.10|$ are significant, $p < .05$, uncorrected. r -values $> |.16|$ are significant, $p < .05$, with a Bonferroni correction for 28 tests.

Predicting FNS and FDS.

Table 4 presents the multiple regression models for FNS and FDS, in each case using age, group (0,1) and their interaction as predictors. Note that the omnibus tests for these models are not always significant, but we are not here concerned with testing a complete model of predictors of FNS/FDS (which are likely to be predicted by many other variables). Our predictions here concern the more specific effects of Color Group and its interaction with age, so the individual tests are relevant. Figure 1 illustrates the interaction between group and age for each variable.

Table 4. Multiple regression predicting FNS and FDS

Criterion: FNS

	<i>b</i>	<i>se</i>	<i>t</i>	<i>p</i>
Color Group	-0.499	3.979	-3.268	0.001
Age	0.023	0.047	0.388	0.698
Age * Color Group	0.509	0.093	3.292	0.001

F=5.62, p=.0009, adj. R-squared = .04

Criterion: FDS

	<i>b</i>	<i>se</i>	<i>t</i>	<i>p</i>
Color Group	-0.371	0.307	-2.398	0.017
Age	-0.049	0.004	-2.398	0.416
Age * Color Group	0.33	0.007	2.103	0.036

F=2.03, p=.109, adj R-squared = .002

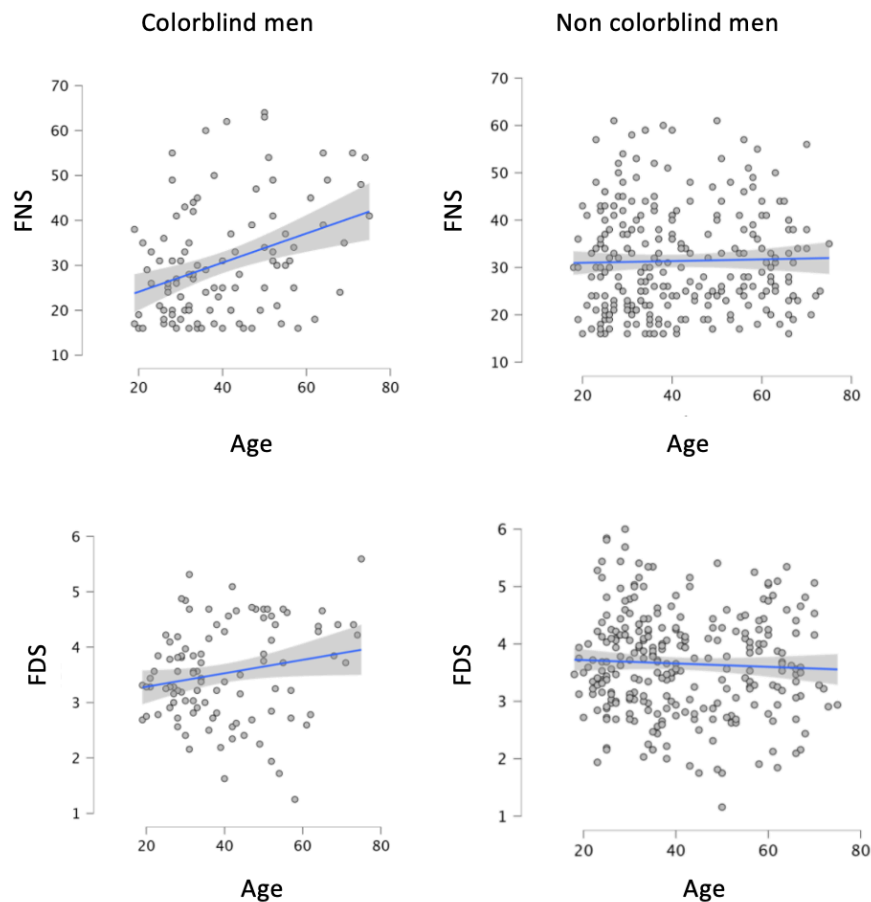


Figure 1. Scatterplots of FNS and FDS as a function of age, in each group (with 95%C.I.).

Figure 1, top panels, illustrates the interaction on FNS. The correlation between age and the FNS was significant for colorblind men ($r = .34, p < .001$) but not for non-colorblind men ($r = 0, p = .96$). Similarly, the bottom panels of Figure 1 shows that the correlation between age and the FDS was significant for colorblind men ($r = .20, p = .04$) but not for non-colorblind men ($r = -.05, p = .39$). Because cross-sectional designs are susceptible to cohort effects (see Siegrist et al., 2013, for a discussion), future work may want to focus on the group difference in younger adults (age ≤ 37 , the median age), whereby FNS scores are lower in colorblind men than in the noncolorblind men (Figure 2, non-colorblind = 28.56, SD = 14.12; colorblind = 24.28, SD = 13.28; $t = 1.90, p = .03, d = .31$). Older men did not differ in FNS as a function of colorblindness (non-colorblind = 28.31, SD = 13.02; colorblind = 31.57, SD = 16.26; Welch's t test = $-1.24, p = .22, d = .22$).

For the FDS in young men, scores were also lower in the colorblind group (Figure 2; non colorblind = 3.75, SD = 0.88; colorblind = 3.46, SD = 0.67; Welch t -test = $2.43, p = .02, d = .37$). Older men did not differ in FDS scores as a function of colorblindness (non-colorblind = 3.57, SD = 0.88; colorblind = 3.61, SD = 1.05; $t = 0.247, p = .81, d = .04$).

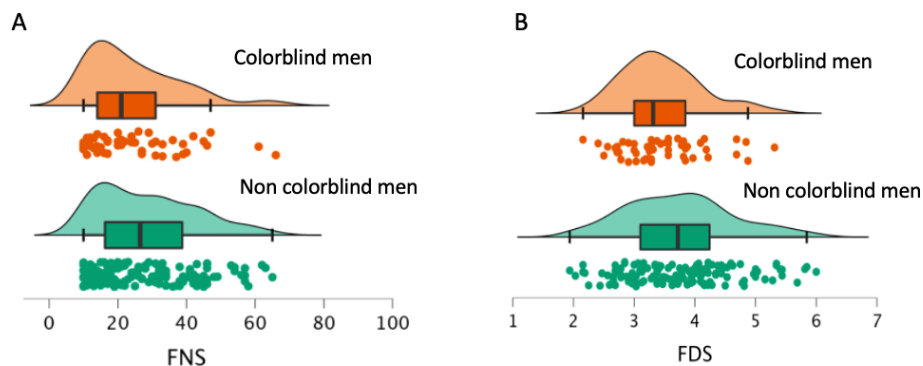


Figure 2. Group difference for FNS (A) and FDS (B) for men 37 years and under.

Predicting Personality Traits.

Table 5 presents the results of regression models using age and group and their interaction for each personality trait. The critical interaction (as well as the group effect) was significant for agreeableness. The interaction observed here for agreeableness is better described as a cross-over interaction, in that agreeableness was higher with age (split according to the median age of 37) in non-colorblind men (mean for young = 3.53, SD= .93; mean for old = 3.73, SD=.98) whereas it was lower as a function of age for colorblind men (mean for young = 3.77, SD= .80; mean for old = 3.49, SD=.93)¹.

Examining Specificity.

The age x color group interaction for FNS and FDS were still significant when controlling for agreeableness in the model (FNS, age x group interaction, $t = 2.832$, $p < .005$; FDS, age x group interaction, $t = 2.104$, $p = .036$).

¹ Because Sun & Gauthier (2023) had not tested the effect of age in their study, we re-analyzed those results and found the age x color group interaction on agreeableness in that study ($t=2.101$, $p=.037$). In that sample, agreeableness did not vary with age in non colorblind men (mean for young = 3.4, SD = .89; mean for old = 3.41, SD = .89) whereas it was lower for older colorblind men (mean for young = 3.83, SD = .71; mean for old = 3.57, SD = .88).

Table 5. Multiple regressions predicting each of the personality traits in the mini-IPIP.

	<i>b</i>	<i>se</i>	<i>t</i>	<i>p</i>
Criterion: Agreeableness				
Color Group	0.31	0.326	1.998	0.046
Age	0.065	0.004	1.083	0.280
Age * Color Group	-0.326	0.008	-2.076	0.039
F=1.44, p=.23, adj. R-squared = .004				
Criterion: Conscientiousness				
Color Group	-0.253	0.284	-1.667	0.096
Age	0.163	0.003	2.790	0.006
Age * Color Group	0.164	0.007	1.064	0.288
F=6.86, p<.001, adj. R-squared = .045				
Criterion: Extraversion				
Color Group	0.136	0.369	0.871	0.384
Age	0.010	0.004	0.167	0.867
Age * Color Group	-0.171	0.009	-1.083	0.280
F=0.519, p=.67, adj. R-squared = -.004				
Criterion: Intellect/Imagination				
Color Group	0.066	0.297	0.423	0.673
Age	0.043	0.004	0.719	0.473
Age * Color Group	-0.107	0.007	-0.681	0.497
F=0.369, p=.78, adj. R-squared = -.005				
Criterion: Neuroticism				
Color Group	-0.124	0.347	-0.818	0.414
Age	-0.218	0.004	-3.722	<.001
Age * Color Group	0.282	0.008	1.837	0.067
F=7.361, p<.001, adj. R-squared = .048				

Study 1b

Method

An open question is why sampling from the population of men on Prolific who report being colorblind results in older samples than when sampling men who report normal color functions on the same platform (this study and Sun & Gauthier, 2023). One likely explanation for 'missing' younger colorblind men is that they are more likely not to be aware that they are colorblind (Birch, 2012). We therefore invited participants from Study 1a to perform a colorblindness screen, to test whether this would affect our conclusions.

Participants

All participants in Study 1a were invited (at most one week after their completion of the first survey) to participate in a study called a “short screening test”. A total of 281 participants (70 of them originally reporting as colorblind) volunteered to participate (83.5% of controls and 68% of colorblind men).

Materials

A 20-trial task was completed online. On each trial, one of the Ishihara plates (Ishihara, 1918) was shown on the screen and participants were asked to choose what they saw among three answers. Two of the plates were anchor plates, expected to be perceived by all. The other 18 trials screened for various degrees of red-green deficiency.

Results

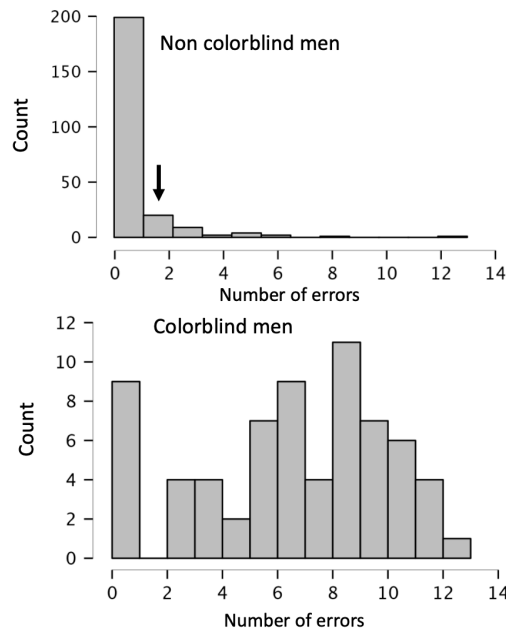


Figure 3. Frequency counts of number of errors on the colorblindness test, in each original group. The arrow indicates cases that were considered “ambiguous”.

Generally validating the test is the fact that average performance on the 18 diagnostic trials in our controls was 98.1% whereas it was 71.1% in colorblind men. However, there were a few individuals who could be considered to be miscategorised. Figure 3 presents the histogram of the number of errors on the test for men reporting as non-colorblind, and for those reporting as colorblind. Nine of the colorblind men made 0 or 1 error and could arguably be re-categorized as having normal color vision.

Additionally, because all colorblind men who made any error made at least 3, we chose 2 errors as an 'ambiguous' level of performance. Twenty men originally reporting no color vision problem were therefore categorized as “ambiguous”, while 22 more were categorized as having some degree of color blindness (making between 3 and 12 errors on the test).

Therefore, approximately 9% of those initially reporting no colorblindness appeared to have some degree of colorblindness (with another 8% ambiguous) while nearly 13% of those reporting colorblindness showed no evidence of it on our test.

We revisited our main analyses using regression to test for the age x group effect we obtained using self-report of colorblindness. We excluded participants who were ambiguous and since the majority of participants had originally correctly reported their status, we decided not to exclude participants who did not return for the colorblindness screening to keep power as high as possible (given our results, we estimate that only another seven participants may switch groups if we had all the data). We computed the regression analyses twice: once including all the switched participants in their new group, and again excluding all the switched participants. There are two possible concerns about the switched participants: one is that they were less conscientious in either not paying attention to the screener question or the colorblindness task, and a second one is that if they were conscientious and truly did not know their colorblindness status, they may not belong in the same group with those who did. In other words, it is possible that any effect of colorblindness on FN and FD may be modulated by knowledge about one's color vision.

Our results provide some evidence that this may be the case, because the critical interaction of group x age on FNS was not significant with the switched participants included (Table 5) but was not when they were not (Table 6). However, we do not have a sufficient number of switched participants to test whether this is a robust difference. The same interaction for FDS, which was already a small effect originally, became non-significant in both recoded samples. Additionally, for both FNS and FDS, the group effect was significant when switched participants were not included, with lower scores in colorblind men.

Table 6. Multiple regression predicting FNS and FDS
with no ambiguous participants but including switched ones

Criterion: FNS

	<i>b</i>	<i>se</i>	<i>t</i>	<i>p</i>
Color Group	-0.278	4.684	-1.787	0.075
Age	0.037	0.062	0.575	0.565
Age * Color Group	0.254	0.109	1.594	0.112

F=2.189, p=.089, adj. R-squared = .01

Criterion: FDS

	<i>b</i>	<i>se</i>	<i>t</i>	<i>p</i>
Color Group	-0.238	0.294	-1.523	0.129
Age	-0.024	0.004	-0.366	0.715
Age * Color Group	0.253	0.007	1.577	0.116

F=0.976, p=.404, adj. R-squared = .000

Table 7. Multiple regression predicting FNS and FDS
with no ambiguous and no switched participants

Criterion: FNS

	<i>b</i>	<i>se</i>	<i>t</i>	<i>p</i>
Color Group	-0.425	4.988	-2.617	0.009
Age	0.002	0.062	0.033	0.974
Age * Color Group	0.418	0.118	2.534	0.012

F=3.184, p=.024, adj. R-squared = .02

Criterion: FDS

	<i>b</i>	<i>se</i>	<i>t</i>	<i>p</i>
Color Group	-0.651	0.313	-2.083	0.038
Age	-0.002	0.004	-0.493	0.622
Age * Color Group	0.014	0.007	1.921	0.056

F=1.587, p=.192, adj. R-squared = .005

General Discussion

We replicated the correlation between color blindness and FN (supporting H1a) and extended it to FD (supporting H1b). In both cases, replicating a pattern observed for FN by Sun and Gauthier (2023), we found an age by group interaction, whereby colorblindness status influenced these traits in younger (less than 38) but not older individuals (supporting H2a and H2b). We verified that these results survived the purification of our sample using a colorblindness test instead of relying on a self-report. We also replicated an unexpected result from Sun and Gauthier (2023), which was an age x group interaction on agreeableness (suggesting that the effects we uncover here are not specific to FD and FN, against H3).

The interaction between age and colorblindness was not originally investigated by Sun and Gauthier (2023) – it was discovered when we reanalyzed their results in preparation for the current work. This interaction, which we replicated here, is difficult to interpret. Both FN and FD have been found to vary with age. FD was positively associated with age in one study (age ranging from 20 to 80+; Egolf et al., 2018; see also Ammann et al., 2018; Oaten et al., 2009) but other work with different disgust measures found a decline with age (e.g., Curtis et al., 2004). In one study with adults 55 and older, age was a positive predictor of FN (van den Heuvel et al., 2019), whereas a large study ($n = 3246$) conducted in Ireland found a peak around 6 years of age, a slow decline between that age and age 17, no change between 17 and 34, and then a monotonic increase with age thereafter (Hazley et al., 2022). Another large study ($n = 4436$, Siegrist et al., 2013) in Switzerland found a small positive correlation between age (ranging between 21 and 99) and FN, but suggested that it was likely a cohort effect. The authors argued that we should expect FN to be reduced with cumulative exposure to a variety of foods, a result they predicted should be obtained in longitudinal studies. As in Sun and Gauthier (2023), we found that

colorblind men reported being more agreeable than noncolorblind men, with an age x group interaction not unlike that observed for FD and FN. Agreeableness is believed to increase slightly and linearly throughout life (Roberts et al., 2006), contrary to the slight decrease with age that we observed in colorblind men. In this case also, a cohort effect is possible and longitudinal work would be necessary to study how these variables change with age. Facing these challenges of interpretation, we advise researchers who wish to pursue these effects outside of a longitudinal approach to limit themselves to a younger adult cohort, in which the effects were most important.

Most important to our current goals is that agreeableness does not appear to account for the FN and FD effects. Although our work is correlational, we have the advantage that colorblindness is not a possible consequence of other traits. But because responses on the FNS and FDS can correlate with aspects of personality, the concern about personality influencing survey responses could also be assuaged in work measuring behaviors predicted by FN and FD, such as food selection and avoidance, as a function of colorblindness. The unexpected relationship between colorblindness and agreeableness may also have important implications for the study of personality. The effects of colorblindness, which is genetic, on personality and attitudes (including FN and FD) could be mediated by a different perceptual experience of the world, or by attributions associated with the knowledge of this difference (for example, explaining some events as being due to colorblindness). But personality traits are in part heritable (Bratko et al., 2017; Sanchez-Roige et al., 2018) and there is some evidence of genetic linkage through pedigree studies between the X-linked colorblindness and affective disorders (Baron et al., 1987; Van Eerdewegh, 1989). FN itself appears to be highly heritable, although this result comes from a study on a sample that was almost entirely female (Knaapila et al.,

2007). Our unexpected results for non food-related aspects of personality may reflect the many possible ways for colorblindness to relate to and/or affect behavior.

The current results, together with those of Sun & Gauthier (2023), are consistent with and extend prior work about the role of color in food processing. Adults' judgments about calorie content and their arousal are predicted by the red/green content of food images (Foroni et al., 2016). But this begs the question of how adults come to use color for food perception.

Developmental studies reveal that children mostly ignore color as a cue for categorization, with the food domain as an exception. For instance, children are more likely to categorize novel objects based on color than shape cues when they believe these items are food (Macario, 1991). However, different work that used unfamiliar but real examples of foods, instead of novel objects, did not replicate this effect (Rioux, Leglaye, et al., 2018). This suggests the familiarity of the stimuli matters, and leads us to highlight the fact that the present work used images of prepared food that was more complex in shape, texture, and color content than most prior work with food that often uses single ingredients on white backgrounds. Our results on the influence of colorblindness on FN and FD suggest that, even if there are innate influences on the role of color in food perception, our experience with colored foods may also play a role.

Constraints on Generality

The most important limitation of this work is that online surveys may include sampling biases confounded with age (Denissen et al., 2010). Our younger participants are more likely to be representative of the broad population, but this is probably not the case as age increases. Much older participants who are not only Internet users but interested in participating in online surveys may not be representative of their age group. For this reason, it could be desirable for

studies following-up on our results to focus resources on the group effect in the adult to mid-life age range. This effect on FN replicates (and extends to FD) and captures our most important predictions. A follow-up study testing participants in all age groups in person or the use of other ways to avoid age-confounded sampling biases may eventually be necessary. Another limitation of this work is that we did not have enough participants whose colorblindness status changed upon testing to ask whether living with colorblindness while being aware vs. unaware is important to the effect on FN and FD.

Methodological and Clinical Implications

One methodological implication of our work is the need to screen for colorblindness across a broad range of studies. Even work that manipulates color to study effects on decisions about food often does not screen for colorblindness (Kennedy et al., 2005; Sun & Gauthier, in press; Suzuki et al., 2017). Our findings show that colorblindness has an impact even in the absence of color manipulation or even when no images are presented, when measuring affective attitudes towards food. Importantly, this effect was obtained even though participants were recruited because they had previously reported being colorblind or not, without knowing about the relevance to this specific study.

Our work may also have clinical implications. In its extreme form, FN can result in malnutrition and psychological difficulties (Rabadán & Bernabéu, 2021; Sarin et al., 2019). Accordingly, there are increasing efforts to develop evidence-based, theoretically informed treatments. A comprehensive cognitive-behavioral approach that includes relaxation training, systematic desensitization with positive imagery and in vivo exposure, cognitive restructuring, and nutritional counseling may be effective (Marcontell et al., 2003). However, the present study suggests that manipulations of perceptual information could be a useful component

of the treatment of eating disorders that share the same spectrum as food neophobia (Dovey, 2018). The present work focused on ‘trait’ color perception. Depending on the mechanism(s) that underlies the associations with FN/FD, implications for treatment could vary. It would be important to measure whether colorblindness also predicts downstream consequences of FN/FD such as food choices. Although colour is generally an important predictor of affective responses to food (Sun & Gauthier, 2023), other visual dimensions may be more important for those who are colorblind and, nevertheless, develop high levels of FN. The desirability of foods is influenced by color in those with high levels of FN (Annette & Stafford, 2023). The present work was motivated by evidence that manipulating ‘state’ color perception (i.e., with grayscale images) can reduce the link between affective responses and perceptual decisions about food. Gauthier and Fiestan (2023) proposed that increasing the perceptual ability to recognize food, as in effective training protocols in other visual domains (Gauthier & Tarr, 1997; McGugin et al., 2011; Wong et al., 2009), could help interrupt the cycle through which FN impacts dietary choices. Importantly, the link between FNS and food recognition ability was assessed during the difficult tasks of perceptual discrimination between visually similar images of prepared food (Gauthier & Fiestan, 2022; Sun & Gauthier, 2023), rather than coarser categorization judgments or naming tasks, as in previous work (see, e.g., Lavin & Hall, 2001; Rioux et al., 2018). Sun & Gauthier (2023) found that the rank-order of individuals in their performance in such judgments was unaffected by removing color, while the correlation of performance to FNS was severed. One (arguably speculative) possibility is that removing color while practicing difficult perceptual discrimination of certain foods could reduce arousal and negative responses (Pliner & Loewen, 2002) while increasing familiarity and perceptual fluency. This experience could then counteract the negative responses even when color is reintroduced. Most treatments to address picky eating

have targeted children (e.g., Pliner & Loewen, 2002; Rioux, Leglaye et al., 2018) but the present approach, targeting fine perceptual skills and complex images of prepared food, could be more appropriate for adults.

Ethics Statement

All procedures in this study were approved by the Institutional Review Board at Vanderbilt Institute [IRB #222116] and conducted in accordance with the ethical standards of the 1964 Helsinki Declaration and later amendments.

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Declaration of competing interest

The authors have no conflicts of interest relevant to this article to disclose.

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Data availability

Data are available on figshare: <https://doi.org/10.6084/m9.figshare.22674622.v1>

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